

(2.1)

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#a.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$		$\sum y_i = 10$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = 1$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = 2$$

#b.

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \times 1 = 1.2$$

The intercept is 1.2, representing the expected value of Y when X is zero.

The slope coefficient is 0.8, indicating that a one-unit increase in X is associated with an average increase of 0.8 units in Y .

#c.

$$\sum_{i=1}^n x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\sum_{i=1}^n x_i y_i = 12 + 4 + 3 + (-1) + 0 = 18$$

$$\Rightarrow \sum_{i=1}^n (x_i - \bar{x})^2 = 10$$

$$\sum_{i=1}^n x_i^2 - 5\bar{x}^2 = 15 - 5 \times 1^2 = 10 \quad \left. \begin{array}{l} \sum_{i=1}^n (x_i - \bar{x})^2 = 10 \\ \sum_{i=1}^n x_i^2 - 5\bar{x}^2 = 10 \end{array} \right\} \Rightarrow \sum_{i=1}^n (x_i - \bar{x})^2 = 10 = \sum_{i=1}^n x_i^2 - 5\bar{x}^2$$

$$\Rightarrow \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 8$$

$$\sum_{i=1}^n x_i y_i - 5\bar{x}\bar{y} = 18 - 5 \times 1 \times 2 = 8 \quad \left. \begin{array}{l} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 8 \\ \sum_{i=1}^n x_i y_i - 5\bar{x}\bar{y} = 8 \end{array} \right\} \Rightarrow \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 8 = \sum_{i=1}^n x_i y_i - 5\bar{x}\bar{y}$$

#d.

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$\hat{y}_i = 1.2 + 0.8x_i$$

$$s_y^2 = \frac{\sum (y_i - \bar{y})^2}{n-1} = 2.5$$

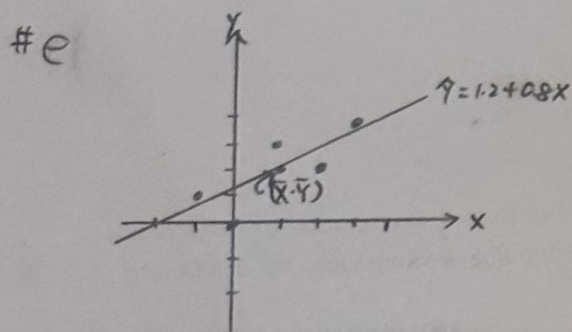
$$s_x^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = 2.5$$

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = 2$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5} \sqrt{2.5}} = 0.8$$

$$CV_x = 100 \times \frac{\sqrt{2.5}}{1} \approx 158.1139$$

The median of X is 1



f. Yes, the fitted line passes through $(\bar{x}, \bar{y})$.

g. $\bar{y} = b_1 + b_2 \bar{x} \Rightarrow 2 = 1.2 + 0.8 \times 1 = 2$ QED

h. $\bar{y} = \bar{y} \Rightarrow \bar{y} = \frac{\sum y_i}{5} = \frac{10}{5} = 2$ } $\bar{y} = \bar{y} = 2$ QED
 $\bar{y} = 2$

#

i. $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{\sum \hat{e}_i^2}{5-2} = \frac{\sum \hat{e}_i^2}{3} = 1.2$

j. $\hat{V}_{b_2}(b_2|X) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{\hat{V}_{b_2}(b_2|X)} = \sqrt{0.12} = 0.346$

(2.22)

(a) Impact of Small Class Size on Total Score

The estimated regression equation is:

$$\text{TOTALSCORE}_i = 916.442 + 12.175 * \text{SMALL}_i$$

- Intercept (916.442): The average total score for students in a regular-sized class is 916.44.
- Coefficient for small (12.175): Students in small classes score, on average, 12.175 points higher than those in regular classes.
- Statistical Significance: The p-value is 0.0236, which is less than 0.05. This indicates that small class sizes have a statistically significant positive effect on total student performance at the 5% significance level.

```
[ (a) 小班 vs 普通班：總分差距 ]
> print(summary(model_a))

Call:
lm(formula = totalscore ~ small, data = data_a)

Residuals:
    Min       1Q   Median       3Q      Max
-189.44  -50.44   -9.44   43.06  324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442     3.675  249.396  <2e-16 ***
small        12.175     5.369    2.268   0.0236 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608, Adjusted R-squared:  0.005323
F-statistic: 5.142 on 1 and 773 DF, p-value: 0.02363
```

(b) Impact on Reading vs. Mathematics Scores

$$\text{READSCORE}_i = 432.665 + 6.924 * \text{SMALL}_i$$

$$\text{MATHSCORE}_i = 483.777 + 5.251 * \text{SMALL}_i$$

- Reading Score: The coefficient is 6.924 with a p-value of 0.0015. This effect is highly significant.
- Math Score: The coefficient is 5.251 with a p-value of 0.1427. This effect is not statistically significant at the 5% level.
- Conclusion: Small class sizes appear to be more effective at improving reading skills than math skills for students.

```

> print(summary(model_b_read))

Call:
lm(formula = readscore ~ small, data = data_a)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665 -21.590  -2.665  16.410  194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665      1.488  290.760 < 2e-16 ***
small         6.924       2.174   3.185  0.00151 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295, Adjusted R-squared:  0.01167
F-statistic: 10.14 on 1 and 773 DF, p-value: 0.001507


> print(summary(model_b_math))

Call:
lm(formula = mathscore ~ small, data = data_a)

Residuals:
    Min       1Q   Median       3Q      Max
-144.777 -35.028  -5.028   29.223  142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777      2.449  197.545 <2e-16 ***
small         5.251       3.578   1.467   0.143
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778, Adjusted R-squared:  0.001488
F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427

```

(c) Impact of Teacher Aides on Total Score

The estimated regression equation is

$$\text{TOTALSCORE}_i = 916.442 + 4.306 * \text{AIDE}_i$$

- Coefficient for aide (4.306): Students in classes with a teacher aide score, on average, 4.31 points higher than those in regular classes without an aide.
- Statistical Significance: The p-value is 0.3888, which is much greater than 0.05.
- Conclusion: There is no statistically significant evidence to suggest that providing a teacher aide improves the total test scores of students.

```
[ (c) 助教班 vs 普通班：總分差距 ]
> print(summary(model_c))

Call:
lm(formula = totalscore ~ aide, data = data_c)

Residuals:
    Min       1Q   Median       3Q      Max
-191.748  -50.442   -5.442   40.558  312.558

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442     3.559  257.529  <2e-16 ***
aide         4.306      4.994   0.862   0.389
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898, Adjusted R-squared: -0.0003068
F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

(d) Impact of Teacher Aides on Individual Subjects

$$\text{READSCORE}_i = 432.665 + 2.871 * \text{AIDE}_i$$

$$\text{MATHSCORE}_i = 483.777 + 1.435 * \text{AIDE}_i$$

- Reading Score: The coefficient is 2.871 with a p-value of 0.161.
- Math Score: The coefficient is 1.435 with a p-value of 0.664.
- Conclusion: Providing a teacher aide does not have a statistically significant impact on either reading or math scores individually.

```
> print(summary(model_d_read))

Call:
lm(formula = readscore ~ aide, data = data_c)

Residuals:
    Min       1Q   Median       3Q      Max
-74.665  -21.536   -2.536   15.464  194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665     1.460  296.341  <2e-16 ***
aide         2.871      2.049   1.401   0.161
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347, Adjusted R-squared:  0.001152
F-statistic: 1.964 on 1 and 835 DF, p-value: 0.1615

> print(summary(model_d_math))

Call:
lm(formula = mathscore ~ aide, data = data_c)

Residuals:
    Min       1Q   Median       3Q      Max
-146.212  -31.212   -5.777   29.223  142.223

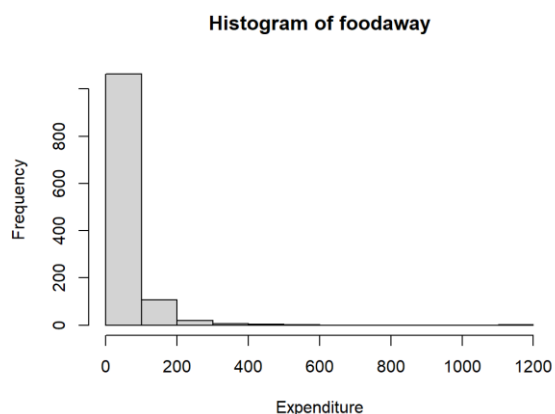
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777     2.355  205.464  <2e-16 ***
aide         1.435      3.304   0.434   0.664
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258, Adjusted R-squared: -0.0009715
F-statistic: 0.1886 on 1 and 835 DF, p-value: 0.6642
```

(2.25)

(a) Summary Statistics for FOODAWAY

- Mean and Median: The average monthly expenditure per person on food away from home is \$49.27, while the median is \$32.55.
- 25th and 75th Percentiles: The 25th percentile is \$12.04 and the 75th percentile is \$67.50.



--- (a) 小題結果 ---

```
> # 將 foodaway 強制轉為數值格式
> cex5_small$foodaway <- as.numeric(cex5_small$foodaway)
> hist(cex5_small$foodaway, main="Histogram of foodaway", xlab="Expenditure")
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00  12.04   32.55   49.27   67.50  1179.00
> quantile(cex5_small$foodaway, probs=c(0.25,0.75))
   25%    75%
12.0400 67.5025
```

(b) Grouped Statistics by Education Level

Expenditure varies notably based on the educational attainment of the household members:

- Advanced Degree: Highest spending with a Mean of \$73.15 and Median of \$48.15.
- College Degree: Moderate spending with a Mean of \$48.60 and Median of \$36.11.
- No Degree: Lowest spending with a Mean of \$39.01 and Median of \$26.02.

```

--- (b) 小題結果 ---
> cat("高級學位 (Advanced):")
高級學位 (Advanced):> summary(subset(cex5_small, advanced == 1)$foodaway)[c("Mean", "Median")]
      Mean      Median
73.15494 48.15000
> cat("大學學位 (College):")
大學學位 (College):> summary(subset(cex5_small, college == 1)$foodaway)[c("Mean", "Median")]
      Mean      Median
48.59718 36.11000
> cat("無上述學位:")
無上述學位:> summary(subset(cex5_small, advanced == 0 & college == 0)$foodaway)[c("Mean", "Median")]
      Mean      Median
39.01017 26.02000
>

```

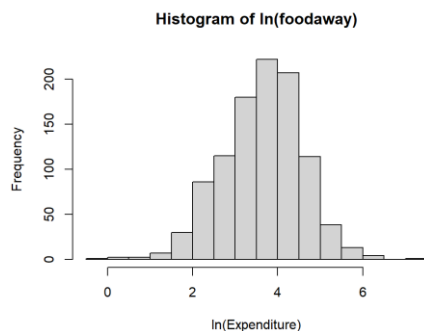
(c) Log-Transformation and Observations

- Observations in foodaway : 1200
- Observations in ln(foodaway) : 1022
- Explanation: The number of observations decreased because some households reported 0 for foodaway. Since the natural logarithm of zero (ln(0)) is undefined (approaching -inf), R excludes these 178 non-positive observations from the log-transformed dataset.

```

--- (c) 小題結果 ---
> # 只有 foodaway > 0 才能取對數，並排除 NA
> data_c <- subset(cex5_small, foodaway > 0 & !is.na(foodaway))
> data_c$ln_foodaway <- log(data_c$foodaway)
> hist(data_c$ln_foodaway, main="Histogram of ln(foodaway)", xlab="ln(Expenditure)")
> summary(data_c$ln_foodaway)
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-0.3011  3.0759   3.6865   3.6508  4.2797   7.0724
> n_a <- nrow(cex5_small)
> cat("Number of observation in foodaway:", n_a, "\n")
Number of observation in foodaway: 1200
> cex5_small$ln_foodaway <- log(cex5_small$foodaway)
> n_c <- sum(is.finite(cex5_small$ln_foodaway))
> cat("Number of observation in ln(foodaway):", n_c, "\n")
Number of observation in ln(foodaway): 1022
>

```



#(d) Linear Regression Estimation

The estimated log-linear regression equation is:

$$\ln(\text{FOODAWAY})_i = 3.1293 + 0.0069 * \text{INCOME}_i$$

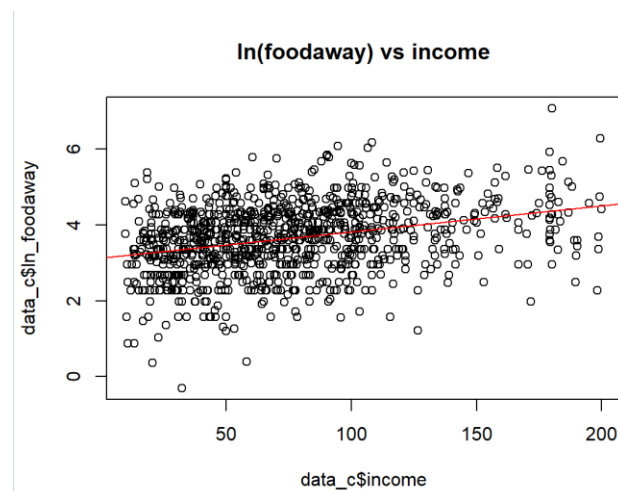
- **Slope Interpretation (beta_2):** This is a log-linear model. The estimated slope is approximately **0.0069**. This implies that for every additional unit of income (which represents **\$100**), the expenditure on foodaway from home is expected to increase by approximately **0.69%** ($0.0069 \times 100\%$).

```

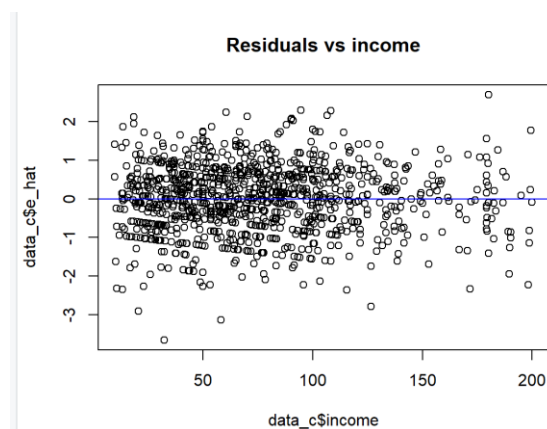
--- (d) 小題結果 ---
> # 確保 income 也是數值
> data_c$income <- as.numeric(data_c$income)
> model_d <- lm(ln_foodaway ~ income, data = data_c)
> coef(model_d)
(Intercept)      income
3.129300436 0.006901748

```

(e) Scatter Plot and Fitted Line



(f) Residual Analysis



- **Patterns:** Upon inspecting the plot of residuals against income, the residuals appear to spread out more as income increases.

- Conclusion: This pattern suggests the presence of heteroskedasticity, meaning the variance of the error term is not constant across different income levels. While the relationship is generally linear, the unpredictability of spending increases for higher-income households.